Solution:

Class	*f*	*cf*
0-4	10	10
4-8	12	22
8-12	8	30
12-16	7	37
16-20	5	42
20-24	8	50
24-28	4	54
28-32	6	60
	N=60	

$$D_5 = \left(\frac{5N}{10}\right)^{th} \text{ item}$$

$$= \left(\frac{5(60)}{10}\right)^{th} \text{ item}$$

$$= 30^{th} \text{ item}$$

This item in the interval 8–12

$l = 8,\ m = 22,\ f = 8,\ c = 4,\ N = 60$

$$D_5 = l + \left(\frac{\frac{5N}{10} - m}{f}\right) \times c$$

$$= 8 + \left(\frac{30 - 22}{8}\right) \times 4$$

$$= 8 + \frac{8}{8} \times 4$$

$$D_5 = 12$$

5.5.3 Percentiles

The percentile values divide the frequency distribution into 100 parts each containing 1 percent of the cases. It is clear from the definition of quartiles, deciles and percentiles

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